

CLAUDE IMPLEMENTATION BRIEF

Offline Forensic RAG — Ingestion Redesign & Hallucination Reduction

Purpose

This document is the implementation instruction for Claude. We are building an offline/local RAG system for mobile forensic data. The most important task is ingestion. The current system can ingest WhatsApp and WhatsApp Business data, but the AI sometimes treats extracted records as generic text, loses their forensic meaning, and hallucinates relationships or conclusions.

Goal: redesign the ingestion representation so that the AI receives structured, provenance-preserving forensic context before embedding and retrieval. Do not replace the source evidence with LLM-generated interpretations.

CRITICAL INSTRUCTION TO CLAUDE

DO NOT immediately rewrite the code.

FIRST:

1. Inspect the existing repository.
2. Inspect the current ingestion pipeline.
3. Inspect current entity/schema definitions.
4. Inspect WhatsApp and WhatsApp Business parsers.
5. Inspect chunking.
6. Inspect embedding generation.
7. Inspect vector database metadata.
8. Inspect retrieval and answer-generation prompts.
9. Run/read existing tests.
10. Identify exactly where hallucination or semantic loss is introduced.

THEN:

Produce a gap analysis against this document.

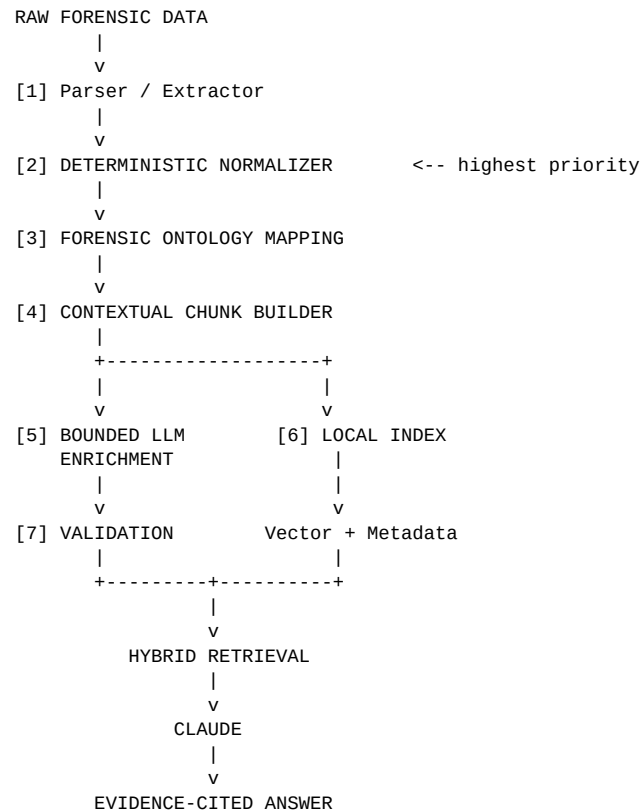
ONLY AFTER the gap analysis:

Propose the smallest safe implementation plan.

Preserve existing working functionality unless there is a clear reason to change it.

Do not invent APIs, fields, tables, parsers, or dependencies without checking the repository first.

1. Target Architecture



2. Core Design Rule

Raw forensic record → direct embedding is NOT acceptable. The canonical evidence representation must be created deterministically first. LLM enrichment is secondary and recomputable.

```
Raw record
-> Canonical forensic object
-> Deterministic forensic context
-> Chunk + metadata
-> Embedding
-> Retrieval
-> LLM answer
```

3. WhatsApp / WhatsApp Business Model

Do not represent WhatsApp as a single generic entity. WhatsApp is the application/source. WhatsApp Messenger and WhatsApp Business should map to a shared messaging ontology while preserving application_variant.

```
application = "whatsapp"
application_variant = "messenger" | "business"
```

Canonical entities:

- Case
- EvidenceSource
- Device
- Application
- Account
- Conversation
- Participant
- Person
- Message
- Attachment
- Call
- ConversationEvent
- TimelineEvent

4. Canonical Message Object

```
{
  "artifact_type": "message",
  "application": "whatsapp",
  "application_variant": "messenger",
  "conversation_id": "CHAT_ID",
  "message_id": "MESSAGE_ID",

  "sender": {
    "platform_id": "PLATFORM_ACCOUNT_ID",
    "resolved_person_id": null,
    "identity_confidence": null
  },

  "receiver_or_target": {
    "platform_id": "TARGET_ID"
  },

  "direction": "incoming",
  "message_type": "text",
  "text": "ORIGINAL MESSAGE",

  "timestamp": "ISO-8601",
  "timestamp_basis": "database_field",

  "conversation": {
    "type": "individual|group|unknown",
    "title": null
  },

  "source": {
    "evidence_id": "EVIDENCE_ID",
    "database": "SOURCE_DB",
    "table": "SOURCE_TABLE",
    "row_id": "SOURCE_ROW"
  }
}
```

5. Deterministic Forensic Context

Generate this from code/templates, not free-form LLM reasoning. It should tell the model what the artifact is, where it came from, who is explicitly identified, and what remains unknown.

FORENSIC ARTIFACT

Artifact Type: WhatsApp Message
Application: WhatsApp Messenger
Conversation Type: Individual Chat
Conversation ID: CHAT_ID

Source Device Account: ACCOUNT_ID
Message Direction: Incoming
Sender Platform ID: SENDER_ID

Timestamp: 20 July 2026 18:30:00 IST
Message Type: Text

Original Content:
"kal milte hain"

PROVENANCE
Evidence ID: EVIDENCE_ID
Database: msgstore.db
Table: message
Row ID: 456

FORENSIC SAFETY
The source-device owner is not automatically the sender.
Sender identity is not established unless supported by evidence.
Do not infer criminal intent or identity from the message alone.

6. Claude Ingestion Enrichment Prompt

You are a forensic data enrichment model.

Input is a NORMALIZED forensic artifact plus deterministic forensic context.

Your task is ONLY to extract semantic metadata directly supported by the input.

STRICT RULES:

1. Never invent people, dates, locations, relationships or events.
2. Never change the original evidence text.
3. Never infer that the device owner is the sender unless explicitly supported.
4. Never infer criminal intent.
5. Never infer identity from a contact name alone.
6. Never turn ambiguous slang into a definitive meaning.
7. Missing information = null.
8. Ambiguity must be explicitly represented.
9. Preserve provenance.
10. Every inference must reference supporting evidence IDs.
11. Return JSON only.
12. Do not create fields outside the approved schema.

Allowed enrichment:

- conversation_type
- communication_topic
- mentioned_people
- mentioned_locations
- mentioned_dates
- mentioned_organizations
- mentioned_phone_numbers
- mentioned_email_addresses
- mentioned_money
- mentioned_urls
- language
- entities_explicitly_mentioned
- semantic_tags
- temporal_expressions
- possible_relationship_signals
- ambiguity_notes

Do not add criminal classifications or unsupported conclusions.

7. Hallucination Example

Input: "Kal maal aa jayega."

Wrong: “Raj is arranging illegal goods.”

Correct: preserve the original text; extract “kal” as a temporal expression; mark “maal” as ambiguous/context-dependent; criminal intent = null.

8. Chunking Strategy

LEVEL 1 – Artifact
One message/event.

LEVEL 2 – Conversation Window
Current artifact + configurable surrounding messages (e.g. ± 5).

LEVEL 3 – Conversation Summary
Participants, date range, message count, major supported topics/events.

LEVEL 4 – Person / Relationship Context
Only relationships supported by evidence.

Do not replace lower-level evidence with summaries.
Summaries are retrieval aids, not evidence.

9. Vector Metadata

```
{
  "chunk_id": "EVID-WA-CHAT-MSG",
  "case_id": "CASE_ID",
  "evidence_id": "EVIDENCE_ID",
  "device_id": "DEVICE_ID",
  "application": "whatsapp",
  "application_variant": "business",
  "artifact_type": "message",
  "conversation_id": "CHAT_ID",
  "message_id": "MSG_ID",
  "sender_id": "P001",
  "receiver_id": "P002",
  "timestamp": "ISO-8601",
  "source_database": "msgstore.db",
  "source_table": "message",
  "source_row": 456
}
```

10. Retrieval Strategy

```
User query
|
v
Case / permission scope
|
v
Exact metadata filters
(app, date, device, sender, conversation, artifact type...)
|
v
Vector search
|
v
Rerank + deduplicate
|
v
Expand context
(parent conversation + participants + nearby timeline events)
|
v
Claude
|
v
Evidence-cited answer
```

11. Identity Rules

Keep separate:

- device owner
- platform account holder
- contact-book name
- message sender
- investigative person/entity

Rules:

- Contact-book names are labels, not verified identity.
- Similar names do not justify merging identities.
- Phone number ownership is not automatically physical device use.
- Device ownership is not proof of authorship of every message.
- Identity resolution must have confidence + supporting artifact IDs.

12. Source of Truth

The canonical evidence store is authoritative. Claude is not. LLM enrichment must be disposable and recomputable. If enrichment is wrong, it must be possible to remove and regenerate it without changing raw evidence or canonical provenance.

13. Final Answer Prompt

You are a forensic evidence analysis assistant.

Answer ONLY from the supplied evidence.

Rules:

- Do not invent missing information.
- Do not establish identity without supporting evidence.
- Do not infer criminal intent.
- Do not merge similar names.
- Do not treat contact-book names as verified identities.
- Do not assume device owner = message author.
- Distinguish observed fact from inference.
- Cite artifact IDs for material claims.
- If evidence conflicts, show the conflict.
- Preserve uncertainty.
- If evidence is insufficient, say:
"The available evidence does not establish this."

For each material conclusion:

FACT:

EVIDENCE:

INFERENCE:

CONFIDENCE:

14. Negative Evidence Rule

No retrieved record is not proof that an event did not occur. The system must say: "No matching artifact was identified in the available/indexed evidence for the specified scope." Never convert absence of retrieval into proof of non-occurrence.

15. Implementation Sequence

Priority	Work
P0	Inspect existing repository and current ingestion path. No code changes yet.
P0	Document current WhatsApp / WhatsApp Business entities and data flow.
P0	Implement/repair deterministic canonical normalization.
P0	Preserve source provenance for every canonical artifact.
P1	Implement deterministic forensic context generation.
P1	Implement artifact + conversation-window chunking.
P1	Add strict vector metadata and exact filtering.
P1	Add bounded Claude enrichment with JSON schema validation.
P1	Add unsupported-claim / provenance validation.
P2	Improve hybrid retrieval, reranking and context expansion.
P2	Add conversation/person/timeline summaries as retrieval aids.
P3	Extend ontology to SMS, calls, Telegram, Signal, browser, gallery, etc.

16. Required Deliverables from Claude

Before implementation:

1. Current architecture map

2. Current schema/entity map
3. Current WhatsApp ingestion flow
4. Hallucination/root-cause analysis
5. Gap analysis against this brief
6. Proposed file/module changes
7. Proposed canonical schema
8. Test strategy

After approval:

9. Implement changes incrementally
10. Add unit tests
11. Add fixture-based WhatsApp tests
12. Test provenance preservation
13. Test ambiguous language
14. Test identity separation
15. Test negative evidence
16. Test retrieval with exact metadata filters
17. Test that unsupported claims are rejected.

17. Definition of Done

The ingestion redesign is complete only when the system can take an actual WhatsApp/WhatsApp Business record and deterministically explain what the record is, where it came from, which conversation/account it belongs to, what is known about sender/direction/time, and what remains unknown—before the record is embedded. Retrieval must be able to recover both the precise artifact and its surrounding conversation context. Claude must answer with evidence IDs and must explicitly refuse unsupported conclusions.

Implementation philosophy: **Evidence first** → **deterministic structure** → **bounded enrichment** → **validated retrieval** → **evidence-cited reasoning**.

This brief is an engineering specification. It does not replace applicable forensic procedures, validation requirements, legal review, or investigator judgment.